

Chronic Disease Prediction System: Predicting Diabetes Mellitus Using Classification Algorithms

Data Mining Project Report
Generated by Gemini

ABSTRACT

Abstract—The early diagnosis and prediction of chronic diseases such as Diabetes Mellitus are critical for effective clinical intervention and patient care. This study proposes a predictive system that analyzes patient physiological features to identify high-risk diabetic individuals. Using the Pima Indians Diabetes Dataset, we comprehensively preprocessed the data by replacing invalid zero values with median column values, followed by standardization. We then trained and evaluated four distinct machine learning classification algorithms: Logistic Regression, Decision Trees, Random Forest, and Gradient Boosting. Performance was evaluated across multiple metrics including Accuracy, Precision, Recall, F1-Score, and ROC-AUC. The Decision Tree model achieved the highest raw accuracy (78.57%), while the Gradient Boosting classifier achieved the highest area under the ROC curve (0.8187), indicating superior separability capabilities. Finally, the best-performing models were utilized in a simulated clinical setting to evaluate a new patient’s risk profile, demonstrating the practical viability of such data mining approaches in modern healthcare.

Keywords—Data Mining, Machine Learning, Diabetes Prediction, Classification Algorithms, Random Forest, Gradient Boosting, Healthcare Informatics

I. INTRODUCTION

Diabetes Mellitus is a chronic, metabolic disease characterized by elevated levels of blood glucose, which over time leads to serious damage to the heart, blood vessels, eyes, kidneys, and nerves. According to the World Health Organization, the global prevalence of diabetes has been steadily increasing, underscoring the urgent need for early detection frameworks. Machine learning and data mining have emerged as powerful tools in the healthcare domain, enabling the extraction of hidden patterns from complex medical datasets.

The primary objective of this project is to develop a Chronic Disease Prediction System specifically targeted at predicting the onset of diabetes. By leveraging physiological and demographic attributes—such as glucose levels, body mass index (BMI), insulin levels, and age—we aim to train robust classification models. A significant challenge in medical datasets is data quality; thus, a rigorous preprocessing pipeline is necessary to handle missing and skewed variables appropriately before any algorithmic training occurs.

In this paper, we present a comparative analysis of four

machine learning models: Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. We examine their strengths and weaknesses using a suite of Key Performance Indicators (KPIs).

II. DATASET AND PREPROCESSING

A. Data Collection

The dataset utilized for this study is the well-known *Pima Indians Diabetes Database*. It comprises diagnostic measurements aimed at predicting whether or not a patient has diabetes based on certain diagnostic measurements included in the dataset. The dataset contains 768 instances and 8 independent variables (Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age) alongside 1 binary target variable (Outcome: 0 for healthy, 1 for diabetic).

B. Data Cleaning and Imputation

In healthcare datasets, missing values are occasionally encoded as ‘0’. Biologically, it is impossible for a patient to have a Blood Pressure or BMI of 0. Analyzing the dataset revealed multiple features with invalid zero values:

- Glucose: 5 missing values
- BloodPressure: 35 missing values
- SkinThickness: 227 missing values
- Insulin: 374 missing values
- BMI: 11 missing values

To prevent model skewness and biases, these zeros were replaced with ‘NaN’ (Not a Number) and subsequently imputed using the median value of their respective columns. Median imputation is robust against outliers, making it highly suitable for medical attributes like Insulin and SkinThickness.

C. Data Scaling and Splitting

Machine learning algorithms, particularly those relying on gradient descent or distance calculations, require features to be on a similar scale. We employed a *StandardScaler* to standardize features by removing the mean and scaling to unit variance:

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

The dataset was split using a stratified approach to preserve the distribution of the target variable, allocating 80% (614 instances) for training and 20% (154 instances) for testing.

III. METHODOLOGY

To identify the most accurate predictive model, we explored both linear and non-linear classification algorithms. Each model's hyperparameters were empirically tuned for optimal performance.

A. Logistic Regression

Logistic Regression is a baseline linear model used for binary classification. It models the probability that a given input belongs to a certain class using the logistic function. The regularization parameter was set to $C = 0.5$ with a maximum of 500 iterations to ensure convergence.

B. Decision Tree

A Decision Tree splits the data based on feature thresholds to maximize Information Gain or minimize Gini Impurity. To prevent overfitting, the tree was constrained with a maximum depth of 4, a minimum of 5 samples required to split an internal node, and a minimum of 2 samples per leaf.

C. Ensemble Methods

Ensemble methods aggregate the predictions of multiple weak learners to form a strong learner.

- **Random Forest:** Operates by constructing a multitude of decision trees at training time. We utilized 200 estimators with a maximum depth of 7.
- **Gradient Boosting:** Builds trees sequentially, with each tree attempting to correct the errors of its predecessor. We configured this classifier with 150 estimators, a learning rate of 0.05, and a subsample ratio of 0.8 to introduce stochasticity and reduce variance.

IV. RESULTS AND DISCUSSION

The models were evaluated on the 20% hold-out test set using five primary KPIs: Accuracy, Precision, Recall, F1-Score, and ROC-AUC (Receiver Operating Characteristic - Area Under Curve).

Table 1: Model Performance Comparison on Test Set

Model	Acc.	Prec.	Recall	F1	AUC
Decision Tree	0.785	0.698	0.685	0.691	0.788
Random Forest	0.753	0.690	0.537	0.604	0.809
Gradient Boost	0.753	0.666	0.592	0.627	0.818
Logistic Reg.	0.694	0.574	0.500	0.534	0.812

As detailed in Table ??, the Decision Tree model exhibited the highest raw accuracy (78.57%) and an excellent balance of Precision (69.81%) and Recall (68.52%), culminating in the highest F1-Score (69.16%).

However, in medical diagnostic scenarios, the ROC-AUC score is often prioritized as it measures the model's ability to discriminate between healthy and diabetic classes across all classification thresholds. Here, **Gradient Boosting** outperformed all others with an AUC of 0.8187, closely followed

by Logistic Regression and Random Forest. This suggests that the ensemble methods yield more reliable probabilistic estimations of patient risk compared to a single Decision Tree.

V. REAL-WORLD APPLICATION SIMULATION

To demonstrate clinical applicability, the Random Forest model was deployed in an inference pipeline to evaluate a simulated new patient. The raw patient metrics were: Pregnancies (2), Glucose (140), BloodPressure (85), SkinThickness (30), Insulin (Missing/0), BMI (31.2), DiabetesPedigreeFunction (0.45), and Age (35).

The pipeline successfully imputed the missing insulin data utilizing the training set's median and scaled the inputs before feeding them to the Random Forest. The system outputted a prediction of **LOW RISK (Healthy)**, with a computed probability of **46.49%** for diabetes onset. Providing probability scores alongside binary outputs gives clinicians a nuanced view of the patient's condition, empowering better preventative care strategies.

VI. CONCLUSION

This study developed and evaluated a Chronic Disease Prediction System using various classification algorithms. Rigorous handling of biologically impossible zeros via median imputation significantly stabilized model training. While the Decision Tree scored highest on standard accuracy, the Gradient Boosting and Random Forest models provided superior class separability (ROC-AUC > 0.80), marking them as highly reliable choices for medical diagnostics. Future work will explore hyperparameter optimization and the inclusion of larger, more diverse demographic variables to further improve the diagnostic certainty of the ensemble models.